



**WEAPONS  
ROBOTICS AND  
CONTROL  
ENGINEERING**  
UNITED STATES NAVAL ACADEMY

HOPPER INFORMATION SERIES

02 / FORCES AND MOTION

# How a quadrotor moves

From four rotor thrusts to vertical, linear and angular motion.

THRUST

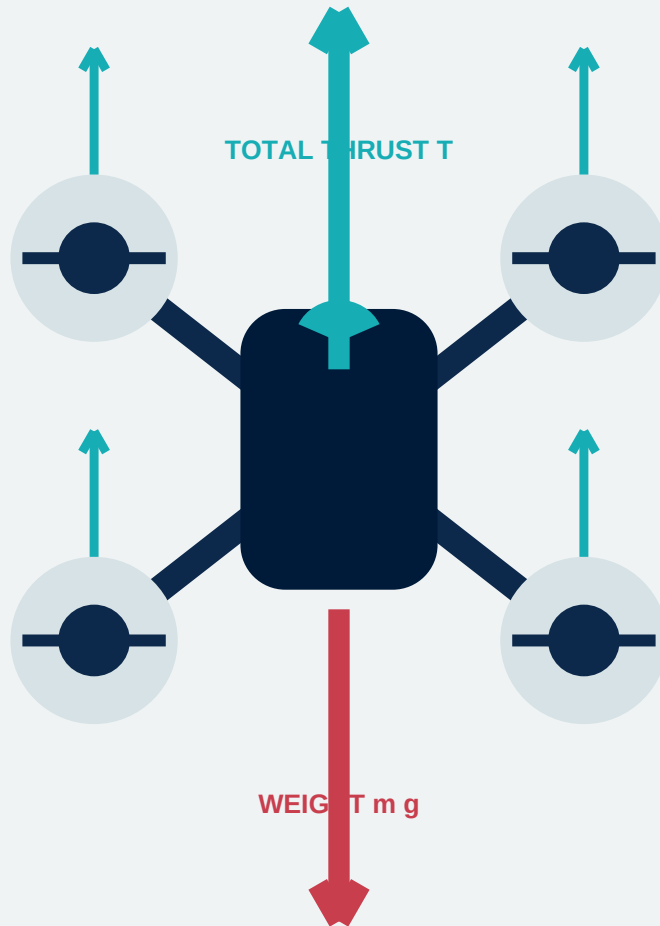
PITCH + ROLL + YAW

SIMPLE MODEL





# Four rotors create one total thrust vector



## HOVER

- All four motors contribute upward thrust.
- When total thrust equals weight, vertical acceleration is zero.
- The flight controller continually corrects small disturbances.

$$T = T1 + T2 + T3 + T4$$

hover:  $T = m g$

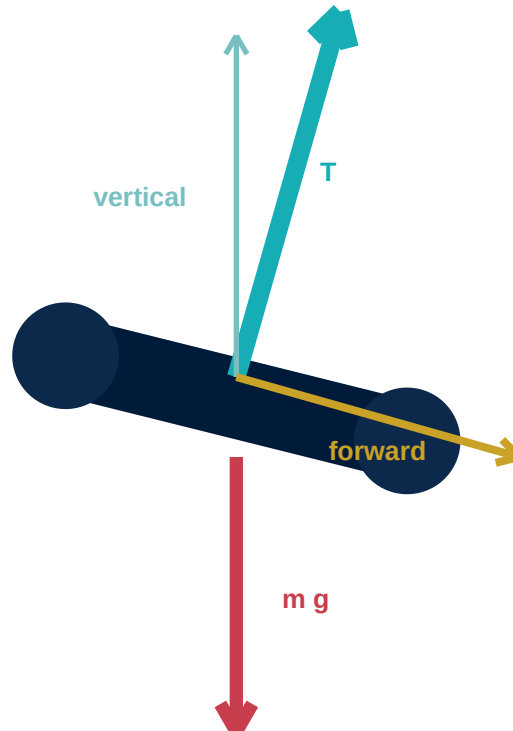
up:  $T > m g$

down:  $T < m g$



# Translation comes from tilting the thrust vector

SIDE VIEW: PITCH FORWARD



The horizontal part of thrust accelerates the aircraft.

## PITCH

Tilt nose forward/backward. Total thrust gains a forward/backward component.

## ROLL

Tilt left/right. Total thrust gains a sideways component.

## UP / DOWN

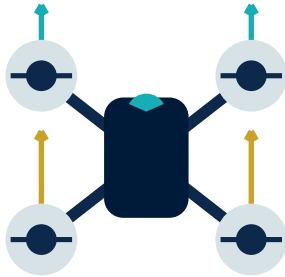
Change total thrust while keeping the body nearly level.



# Angular motion comes from unequal rotor forces

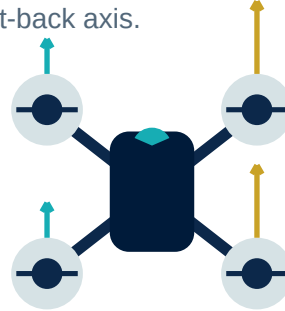
## PITCH TORQUE

More thrust on one fore/aft side rotates the body about its left-right axis.



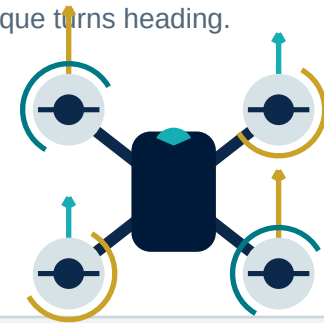
## ROLL TORQUE

More thrust on one left/right side rotates the body about its front-back axis.



## YAW TORQUE

Shift power between counter-rotating rotor pairs; reaction torque turns heading.



ROLL  $\phi$



PITCH  $\theta$



YAW  $\psi$



Angular acceleration follows net torque, not simply motor speed.



# Motor mixing at a glance

| COMMAND | ROTOR CHANGE                         | RESULT              |
|---------|--------------------------------------|---------------------|
| Up      | All four increase together           | More total thrust   |
| Down    | All four decrease together           | Less total thrust   |
| Pitch   | Fore/aft pair difference             | Nose tilts          |
| Roll    | Left/right pair difference           | Body tilts sideways |
| Yaw     | One spin pair up; opposite pair down | Heading rotates     |

The flight controller performs this mixing continuously; student code requests motion, not raw motor RPM.



# A very simple mathematical model

## TOTAL THRUST

$$T = T_1 + T_2 + T_3 + T_4$$

## VERTICAL

$$m \ddot{z} = T \cos(\phi) \cos(\theta) - m g$$

## NEAR HOVER

$$\ddot{x} \sim g \theta \quad \ddot{y} \sim -g \phi$$

## ROTATION

$$I \alpha = \tau$$

## WHAT THE SYMBOLS MEAN

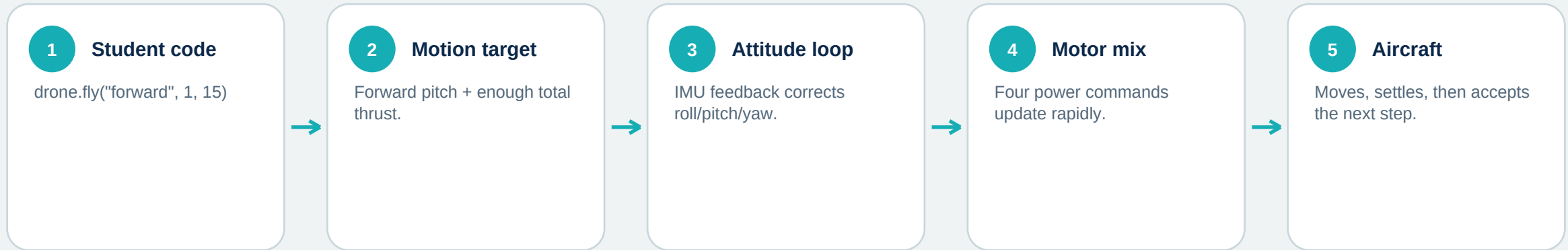
- $m$ : mass;  $g$ : gravitational acceleration.
- $\phi$  /  $\theta$ : roll and pitch angles.
- $I$ : rotational inertia;  $\tau$ : net torque.
- $\alpha$ : angular acceleration.

## ASSUMPTIONS

- Rigid aircraft; calm indoor air.
- Small roll/pitch angles near hover.
- Motor and propeller dynamics simplified.
- Axis signs depend on the chosen coordinate frame.



# From code to stabilized motion



```
HOPPER PROGRAM.JS

1  await drone.takeOff();
2  await drone.fly("forward", 1, 15);
3  await drone.hover();
4  await drone.land();
```

## WHY SEQUENCING MATTERS

- await keeps commands in order.
- Timed fly and rotate commands are open-loop motion requests.
- Built-in settle time reduces overlap between maneuvers.
- Vision checks should happen after the aircraft stabilizes.



# Teach the model, then test the model

## QUICK DEMONSTRATIONS

- Hold a paper quadrotor level: where does thrust point?
- Tilt it 10 deg: what horizontal force appears?
- Compare all-motor change with paired-motor change.
- Predict motion before running a 15% simulator command.

## SOURCES AND FURTHER READING

### NASA STEM - The Science Behind Quadcopters

<https://www.nasa.gov/stem-content/the-science-behind-quadcopters/>

### Stanford - Quadrotor Helicopter Flight Dynamics and Control

[https://ai.stanford.edu/~gabeh/papers/Quadrotor\\_Dynamics\\_GNC07.pdf](https://ai.stanford.edu/~gabeh/papers/Quadrotor_Dynamics_GNC07.pdf)

### Hopper Studio - classroom simulator dynamics

lib/simulation.ts in the Hopper Studio project

**PREDICT -> COMMAND -> OBSERVE -> EXPLAIN**

The simple equations are a reasoning tool, not a precision flight model.